CSE 471

System Analysis and Design



COURSE COORDINATOR

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COURSE MATERIALS

Resources

T

Text book

System Analysis and Design, 5th edition — Dennis, Wixom, Roth.

S

Presentation slides

Lecture slides provided throughout the course.

+

Other resources

Other resources will be shared between the lecture.

LECTURE 01

Today's Roadmap

1.0

SEGMENT 1

What is System analysis and design

2.0

SEGMENT 2

Software lifecycle

3.0

SEGMENT 3

DevOps

4.0

SEGMENT 4

DevOps (cont.)

5.0

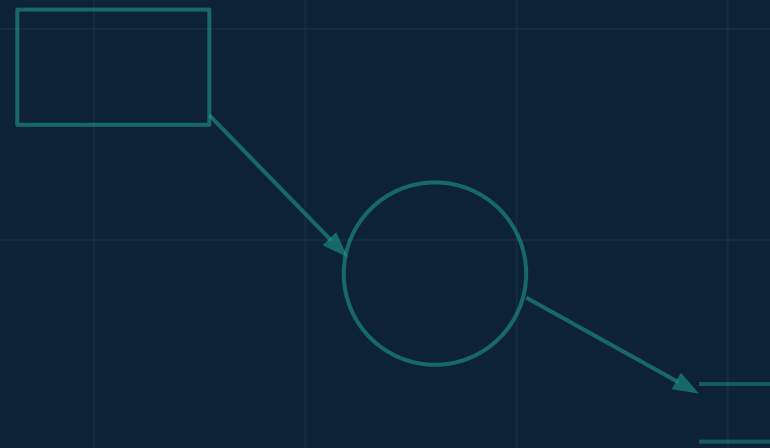
SEGMENT 5

Project Team Roles and Skills

1.0

SEGMENT 01

What is System Analysis and Design



1.0

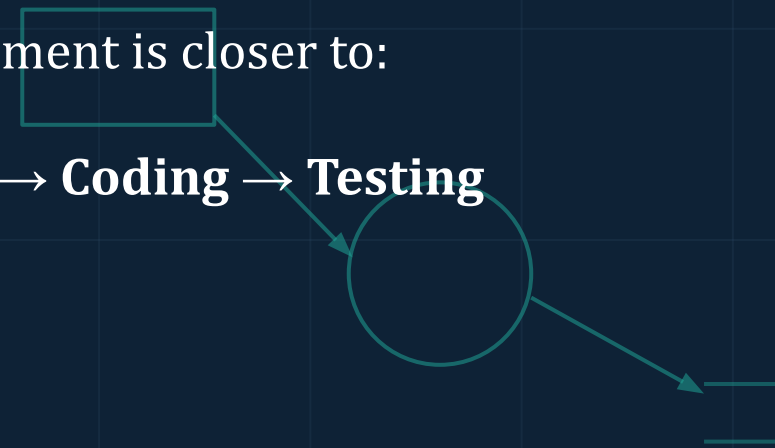
Analysis → What does the system need to do?
Design → How are we going to build it?

- Students often think software development means:

  Requirement → **Coding**

- But this course teaches them that good software development is closer to:

  Problem → Analysis → Requirements → Design → Coding → Testing → Deployment



SEGMENT 1

Introduction

1.0

The SDLC

The systems development life cycle (SDLC) is the **process of determining how an information system (IS) can support business needs, designing the system, building it, and delivering it to users**.

2.0

The systems analyst

The systems analyst plays a key role in the SDLC, analyzing the business situation, identifying opportunities for improvements, and designing an IS to implement the improvements.

3.0

The primary goal

The primary goal of the system analyst is to create value for the organization, which for most companies means increasing profits.

4.0

Change agents

Systems analysts do things and challenge the current way that an organization works.

Introduction — Continued

2%

of $100M+ projects
are successful

— Standish Group analysis

1.0

Susceptible to failure

Large systems development projects are particularly susceptible to failure.

2.0

The evidence

An analysis of very large development projects conducted by the Standish Group found that for projects that exceed $100 million in labor costs, only 2% are successful.

3.0

Major factors

Several major factors were — **delays in decision making and a high workforce turnover** during the project.

Information System

1.0

What they are

Information systems are software applications which manage large amounts of data, share information etc.

2.0

Most software is IS

Most of the software out there is information systems software,

Information System vs. Software

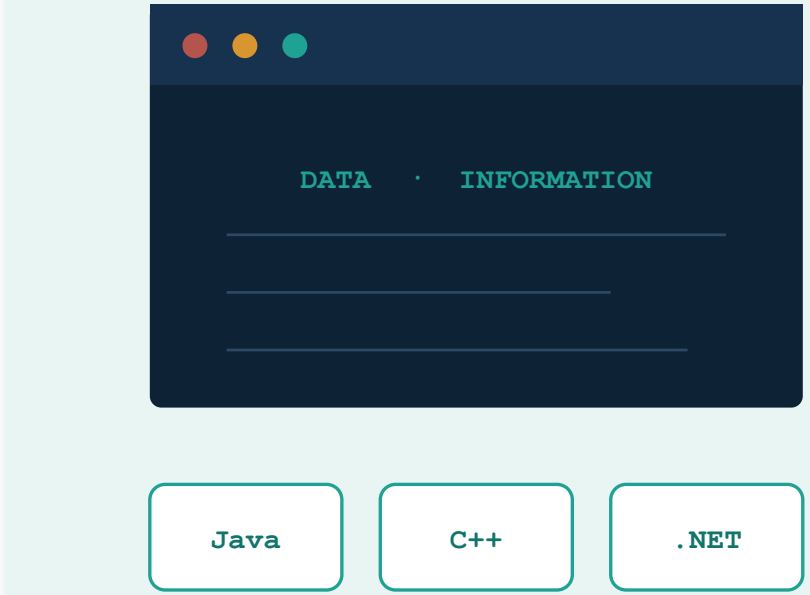
- Software: The application or program itself.
- Information System: The complete arrangement of software, hardware, people, data, and processes working together.

Remember: Software is often a component of an Information System, but an Information System is more than just software.

3.0

Built with mainstream languages

written in languages such as Java, C++, .NET and the like.



Software that manages & shares data

SEGMENT 1

Components of IS

HW

Hardware

These are the devices like the monitor, processor, printer and keyboard, all of which work together to accept, process, show data and information.

SW

Software

The programs that allow the hardware to process the data.

DB

Databases

The gathering of associated files or tables containing related data.

NW

Networks

A connecting system that allows diverse computers to distribute resources.

PR

Procedures

The commands for combining the components above to process information and produce the preferred output.

WHEN SYSTEMS FAIL

Software Horror Stories



Bank of America

$1B+

Spent $23,000,000 on a 5-year project to develop a new accounting system. Spent over $60,000,000 trying to make the new system work, finally abandoned it. Loss of business estimated in excess of $1,000,000,000.



The B1 Bomber

+$1B

Required an additional $1,000,000,000 to improve its air defense software, but the software still isn't working to specification.



Ariane 5, Flight 501

$500M

The loss of a $500,000,000 spacecraft was ultimately attributed to errors in requirements, specifications and inadequate software reuse practices.

WHEN SYSTEMS FAIL

The Bad News

30%

of large IT projects are cancelled before completion

50%

of IT projects are over-budget by more than 200%

≤60%

of prescribed functionality is delivered by the majority of completed projects

4.0

Under-used systems

Many delivered information systems are under-used because they don't meet user needs and/or expectations.

5.0

Legacy bottleneck

Legacy systems are a serious and growing bottleneck to organizational evolution.

MOTIVATION

Why is this Course Important?

54%

Most errors are detected after coding and testing

45%

Almost half of all errors in software are in requirements and design

77%

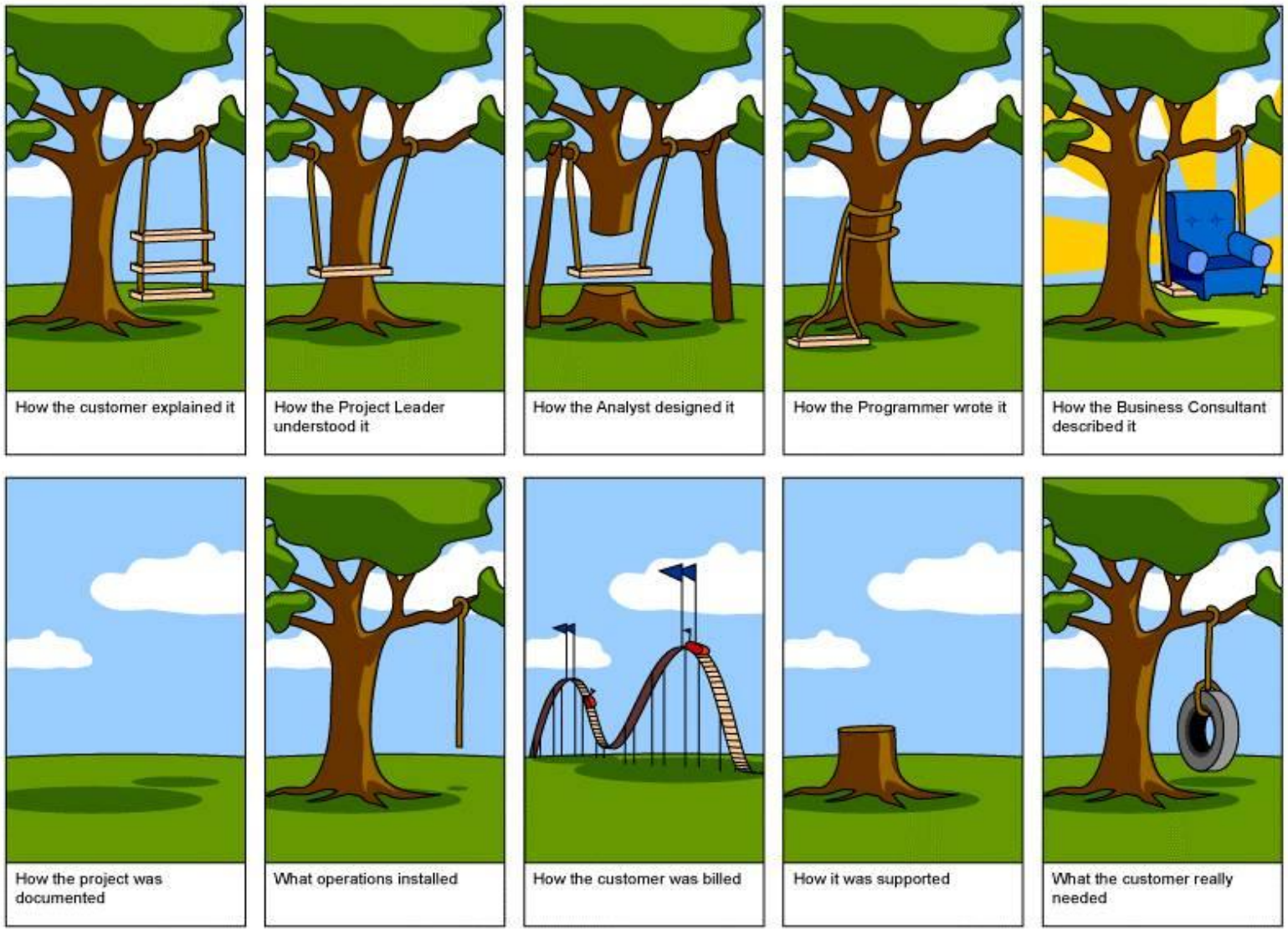
Most errors made during requirements analysis are non-clerical

100×

Requirements errors can cost up to 100 times more to fix than implementation errors, if not caught early on

→ **Need to do requirements and design right!**

WHAT COULD POSSIBLY GO WRONG



Key Ideas

✗

Why systems fail

Many failed systems were **abandoned** because analysts tried to build wonderful systems without understanding the organization.

✓

The primary goal

The primary goal is to create value for the organization.

Key Ideas

Systems analyst is a key person

1.0

Analyze

analyzing the business

2.0

Identify

identifying opportunities for improvement

3.0

Design

designing information systems to implement these ideas

SEGMENT 1

The Systems Analyst

1.0

Team player

The systems analyst works closely with all project team members so that the team develops the right system in an effective way.

2.0

Technology applier

Systems analysts must understand how to apply technology to solve business problems.

3.0

Change agent

Systems analysts serve as **change agents who identify the organizational improvements needed, design systems to implement those changes, and train and motivate others to use the systems.**

Systems Analyst Skills

</>

Technical skills

Analysts must have the technical skills to understand the organization's **existing technical environment**, the new system's technology foundation, and the way in which both can be fit into an integrated technical solution.

$

Business skills

Business skills are required to understand how IT can be applied to business processes and to ensure that IT delivers real business value.

?

Analytical skills

Analysts are continuous problem solvers at both the project and the organizational level, and they put their analytical skills to the test regularly.

What is System Analysis?

1.0

Definition

The collection of **notations, methodologies and tools** used to gather details and analyze a problem situation prior to information system design and implementation.

2.0

What it must ensure

Systems analysis (or, requirements analysis) must ensure that the proposed information system meets user needs, can be delivered on time, and can be updated inexpensively.

3.0

Where it goes wrong

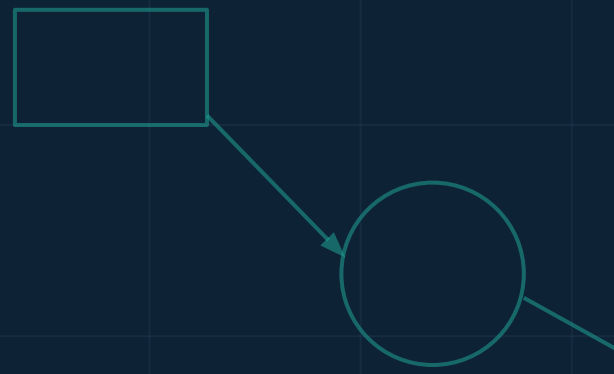
Problems in "getting the systems analysis right", such as ill-defined situations, ambiguities, inconsistencies, mixing requirements with design.

Need for Systems Analysis?

REMEMBER

100×

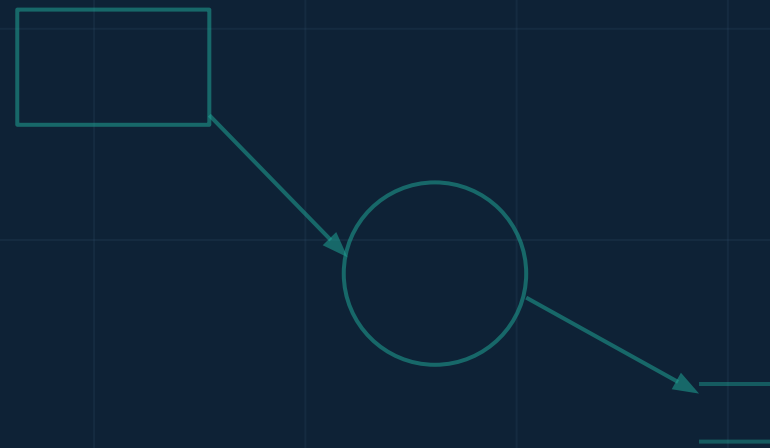
Finding and fixing a fault after software delivery is 100x more expensive than finding and fixing it during systems analysis or early design phases.



2.0

SEGMENT 02

Software Lifecycle



Major Attributes of the Life Cycle

The project…

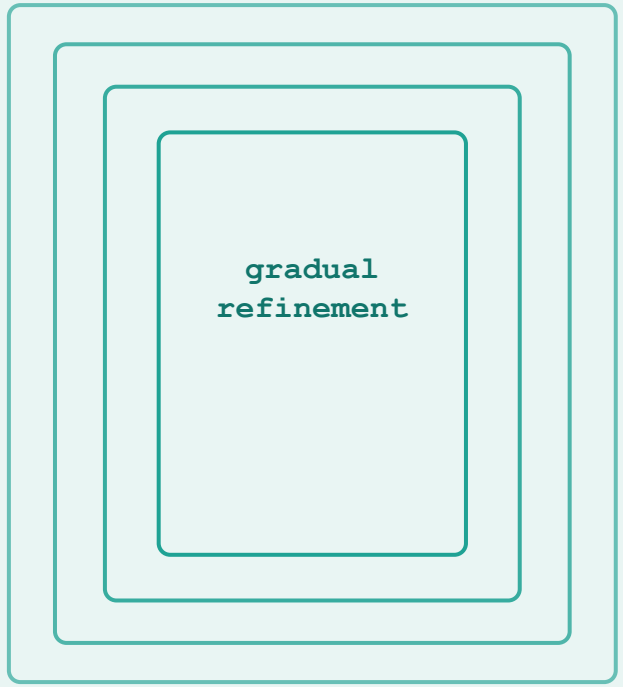
1.0 Moves systematically through phases where each phase has a standard set of outputs

2.0 Produces project deliverables

3.0 Uses deliverables in implementation

4.0 Results in actual information system

5.0 Uses gradual refinement



SEGMENT 2

Project Phases

1

Planning

Why build the system?

2

Analysis

Who, what, when, where will the system be?

3

Design

How will the system work?

4

Implementation

System delivery

Idea → Planning → Analysis → Design → Implementation → System success

Planning



1.0 Identifying business value

2.0 **Analyze feasibility**

3.0 Develop work plan

4.0 Staff the project

5.0 Control and direct project

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Analysis Phase

1.0 Determine analysis strategy

→ *Study existing system and its problems*

2.0 Collect and analyze requirements

→ *Develop new system concept · Describe new system with analysis models*

3.0 Prepare and present system proposal

→ *Summarize results of the Analysis Phase · Go/No Go decision made by sponsor and steering committee*

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Design Phase

1.0 Determine design strategy

→ *Build / buy / outsource*

2.0 Design system components

→ *Architecture, interface, database, programs*

3.0 Assemble design elements into system specification

4.0 Present to steering committee

→ *Go / no go decision before entering final phase*

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Implementation Phase

1.0 **System construction**

→ ***Programming and testing***

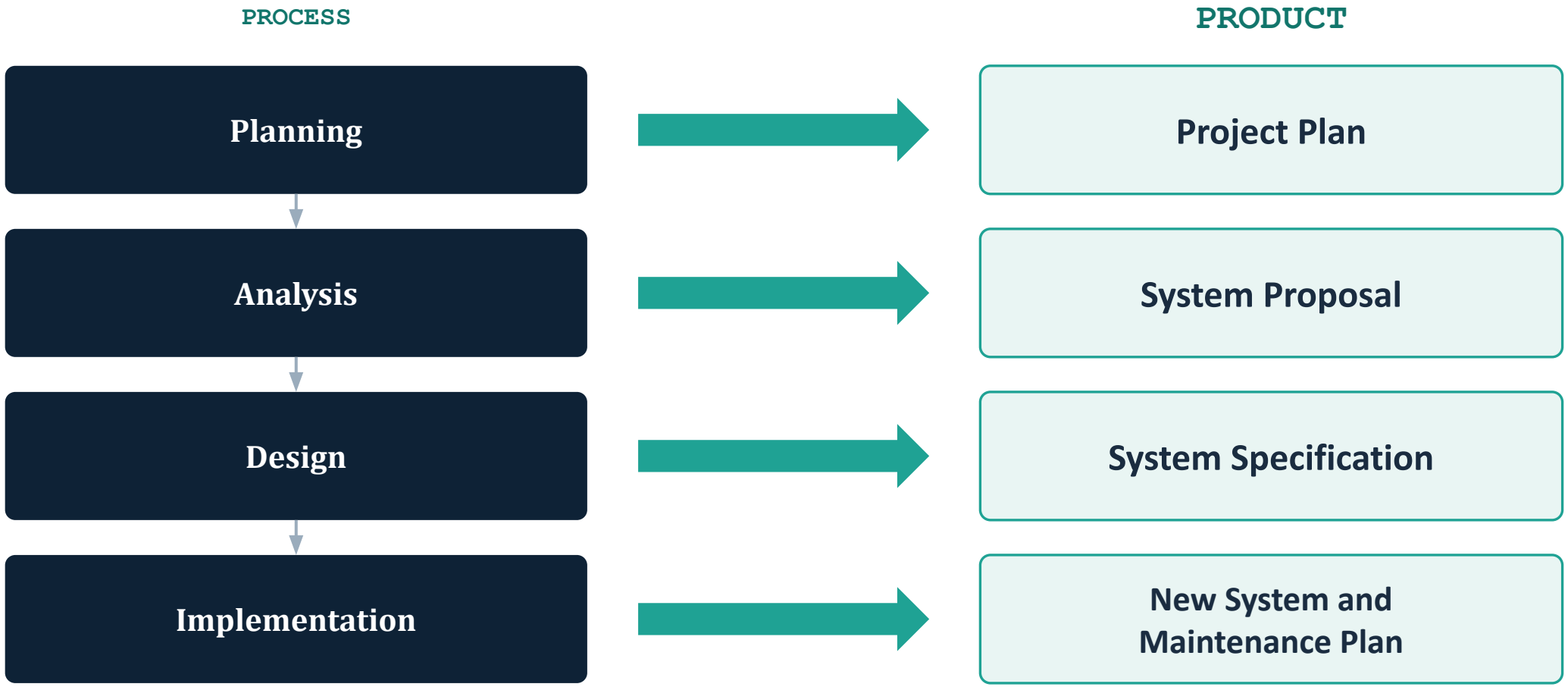
2.0 **System installation**

→ ***Training · Conversion to new system***

3.0 **On-going system support**

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Processes and Deliverables



SEGMENT 2

What is a Methodology?

A formalized approach or series of steps to implement SDLC

Methodology categories:

1.0 **Process-centered**

2.0 **Data-centered**

3.0 **Object-oriented**

SEGMENT 2

Need for Methodology

✗

Code-first rarely scales

Writing code without a well-thought-out system request may work for small programs, but rarely works for large ones.

✓

Design before building

Need to have a good design from the requirements before moving on to implementation.

Systems Development Methodologies

1 Structured Design

- Waterfall Development
- Parallel Development

2 Rapid Application Development (RAD)

- Phased Development
- Prototyping

3 Agile Development

- Extreme Programming

4 DevOps

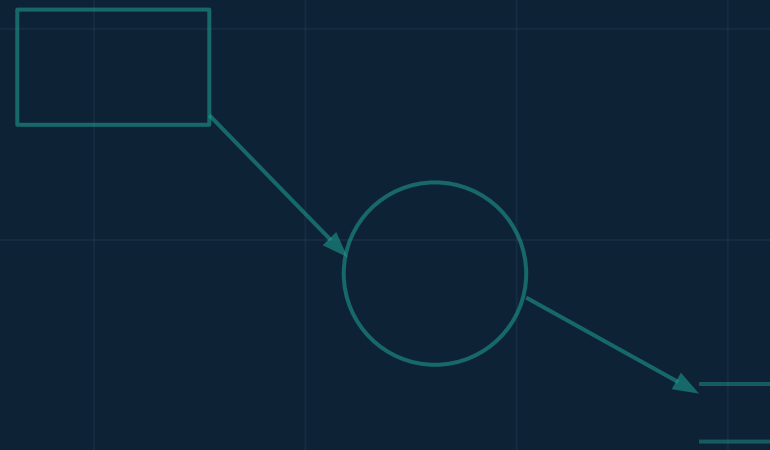
Covered in Segments 3 & 4

Extreme Programming (XP) is an Agile method where software is developed in **small, frequent releases**, with **continuous feedback, testing, and close teamwork**.

3.0

SEGMENT 03

DevOps



Current SDLC Methodology

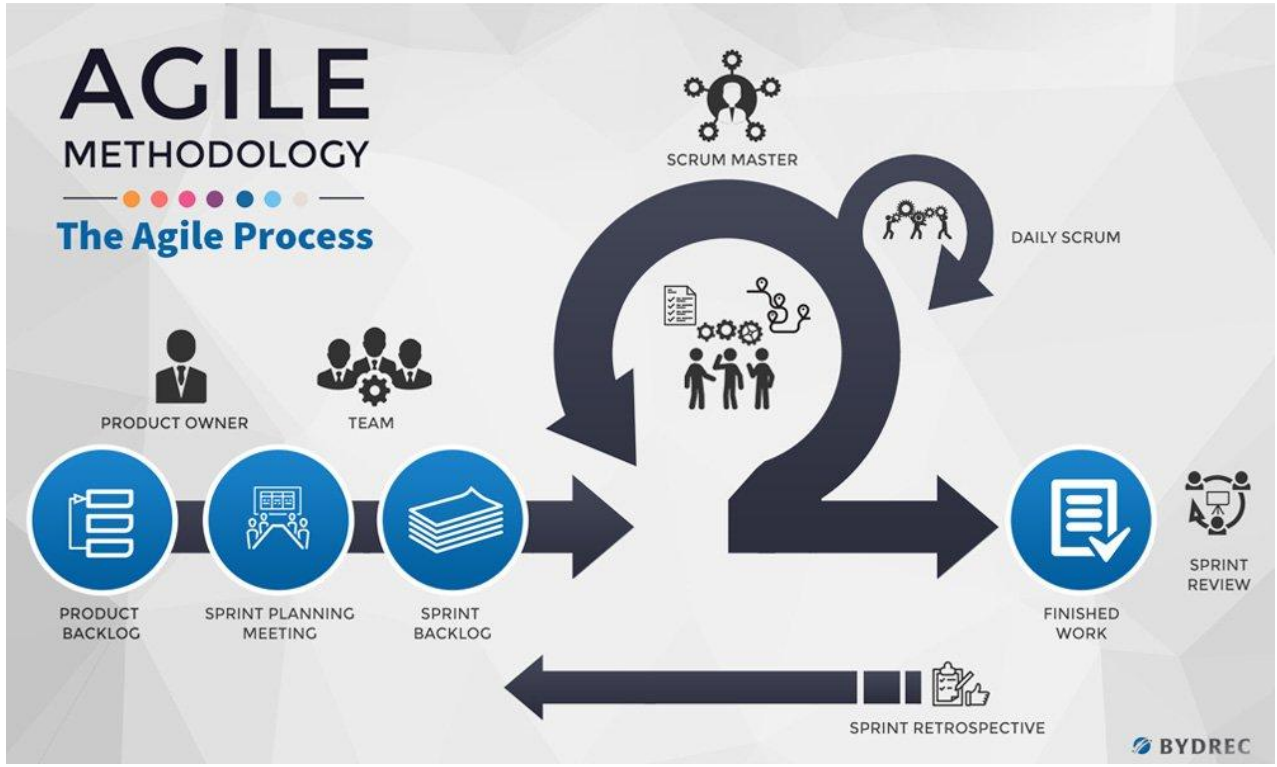
Agile = a philosophy / way of working
Scrum = a specific framework used to apply Agile

1.0 **Waterfall**

2.0 **Prototyping**

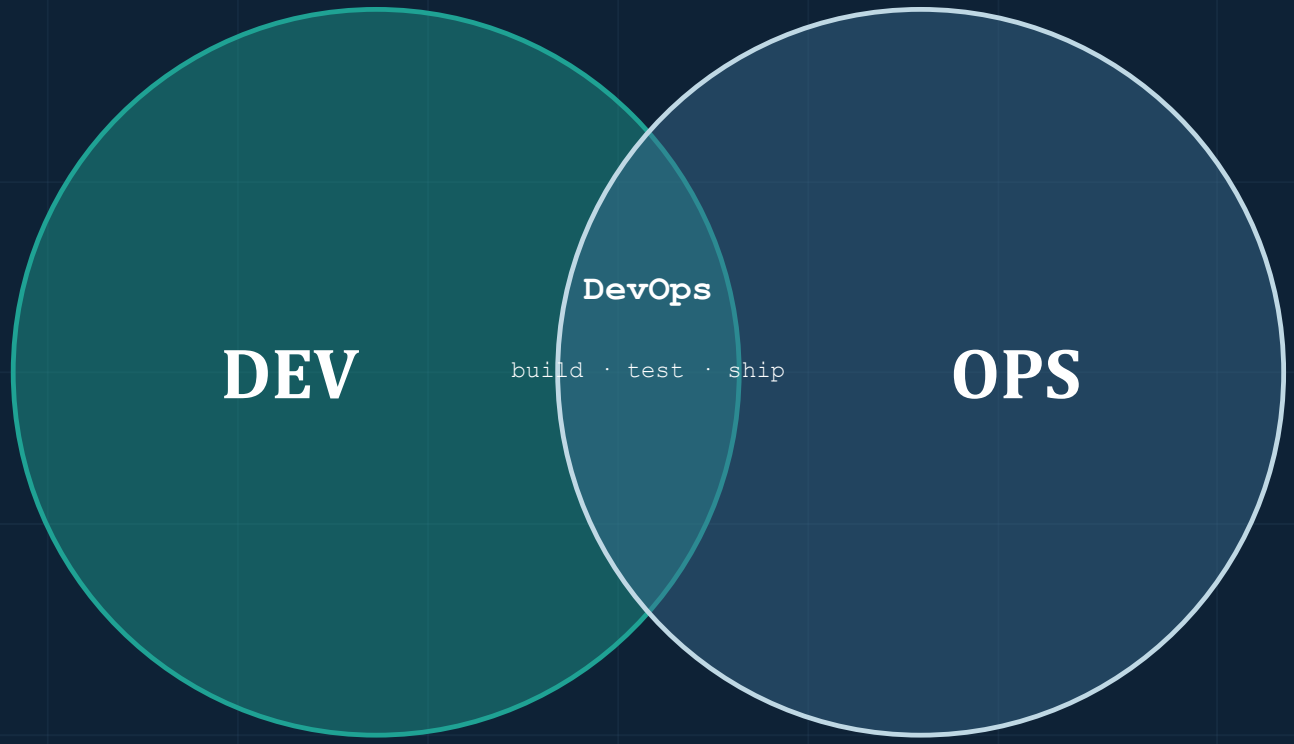
3.0 **Agile**

4.0 **DevOps**



What is DevOps?

DevOps is a combination of cultural philosophies, practices, and tools that unites software development (Dev) and IT operations (Ops) into a single, continuous process



Development and Operations — no longer working apart

SEGMENT 3

Why DevOps?

1.0

Isolated teams

Before DevOps, operation and development teams were working in an isolated environment.

2.0

Late, slow testing & deployment

Testing and Deployment activities mostly were performed in an isolated manner after design-build step, and they took more time than actual project completion time.

3.0

Time drain

Team members usually spend a large amount of time in deploying, testing, designing, and building the projects.

SEGMENT 3

Why DevOps?

4.0

Manual errors

Human production errors were deployed during manual code conduction.

5.0

Out-of-sync timelines

Operations and coding teams generally had different timelines and did not have proper synchronization that results in further delay.

6.0

The urgent need

To avoid the hassles mentioned above and non-collaborative performance measures, there was an urgent need for robust IT technology like DevOps to satisfy business owners and stakeholders.

SEGMENT 3

Current Relation Among Development and Operation Team

DEV

Develop

The development team worked on the software, developing it and making sure that the code worked perfectly.

→

Hand over

After hours of hardwork and a lot of trial and error, the team releases a code which has to be executed by the Operations team which is responsible for the release and operation of the code.

OPS

Operate & report

The operations team will be checking the application and its performance and reporting back any bugs, if present.

What DevOps Can Do?

1.0

Bridge the gap

To bridge this gap, Development ('Dev') team and Operations ('Ops') team collaborated giving rise to DevOps.

2.0

Containerize

The Development team encapsulated their code in a container which is a lightweight software environment.

3.0

Hand off & run

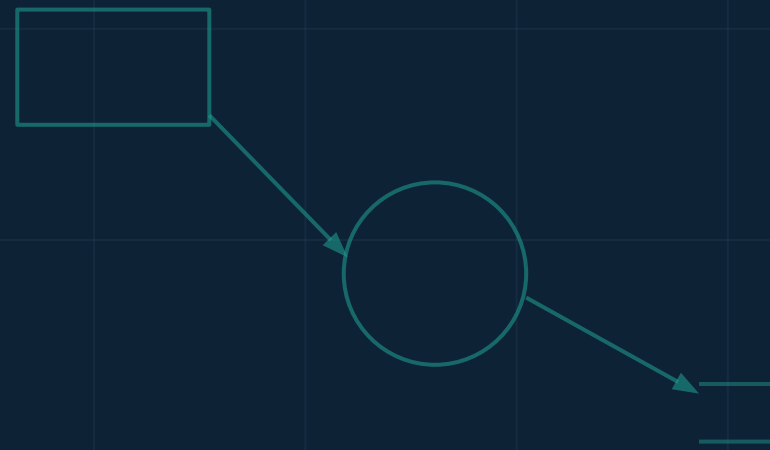
When the developers were done with their work, they would simply pass on this container along with the code to the operations team. The Ops will run this container, along with the code, and it worked as expected!



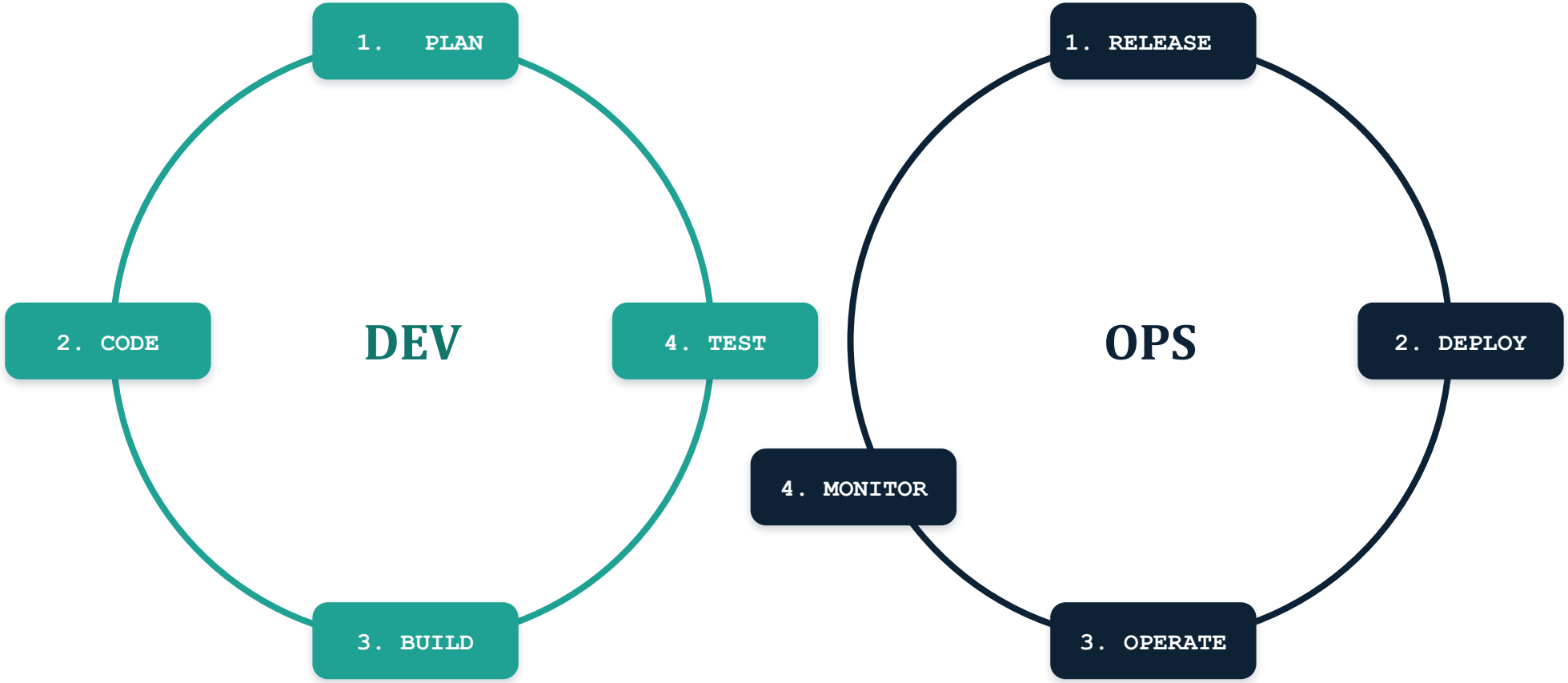


SEGMENT 04

DevOps (cont.)



Steps of DevOps



Steps of DevOps

PLAN

The planning stage gathers information from the monitoring stage and, as per feedback, implements the changes for better performance.

CODE

The first step in the DevOps lifecycle is coding, where developers build the code on any platform.

BUILD

Developers build the version of their program in any extension depending upon the language they are using.

TEST

For DevOps to be successful, the testing process must be automated using any automation tool like Selenium.

Steps of DevOps

RELEASE

A process for managing, planning, scheduling, and controlling the build in different environments after testing and before deployment.

DEPLOY

This phase gets all artifacts/code files of the application ready and deploys/executes them on the server.

OPERATE

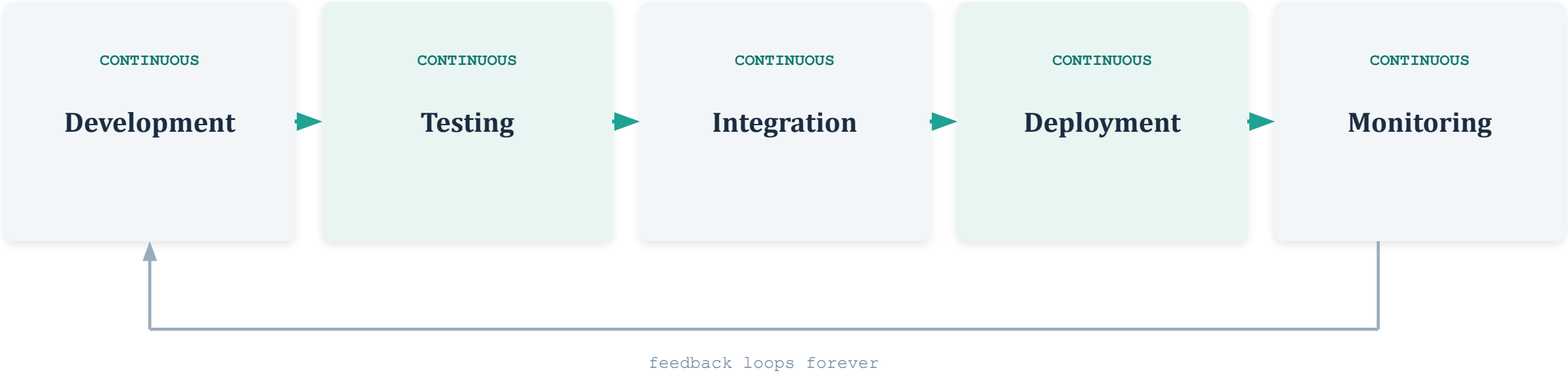
The application is run after its deployment, where clients use it in real-world scenarios.

MONITOR

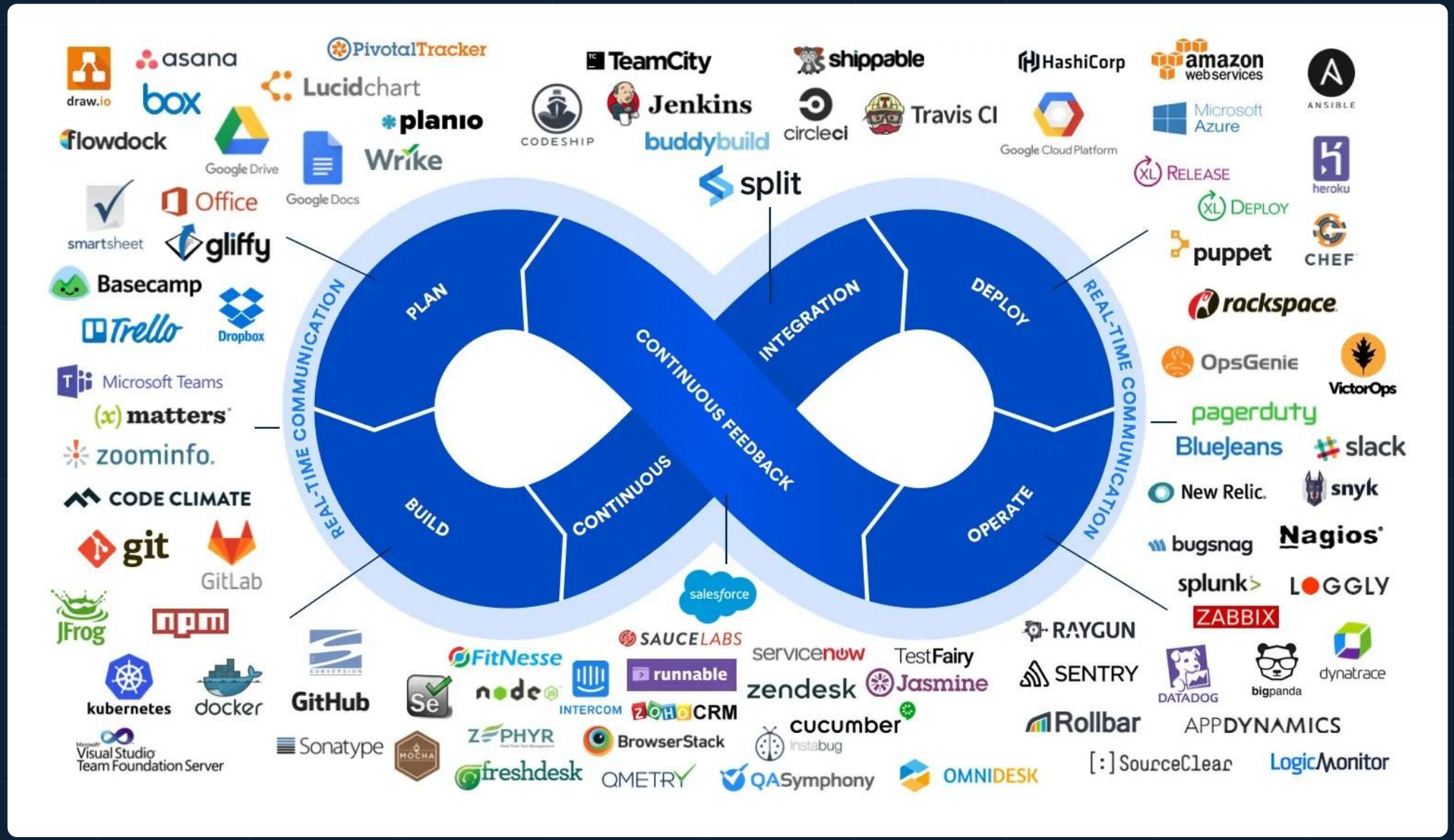
This phase helps in providing crucial information that basically helps ensure service uptime and optimal performance.

SEGMENT 4

Everything Should Be Continuous



THE DEVOPS TOOL LANDSCAPE



The Advantage of DevOps

Example: Food Delivery App

Imagine you own a food delivery app.

Without DevOps

A customer reports:

> "Order tracking is not working."

The company waits 2 months for the next release.

Result:

- Customers get frustrated.
- Bad reviews increase.
- Revenue decreases.

The Advantage of DevOps

With DevOps

1. Developer fixes the bug.
2. Automated tests run immediately.
3. Code is integrated automatically.
4. New version is deployed automatically.
5. Monitoring checks if the fix works.
6. Feedback is collected.

The bug may be fixed within hours instead of months.

- Faster updates
- Fewer bugs
- Better customer satisfaction

The Advantage of DevOps

Real-Life Example: Facebook/Instagram

When Facebook adds a new feature:

- Developers write code.
- Automated tests run.
- Code is integrated.
- Feature is deployed.
- Usage is monitored.
- Feedback is collected.

This cycle repeats continuously.

SEGMENT 4

The Advantage of DevOps

- ✓ Reduced chance of product failure
- ✓ Improved flexibility and support
- ✓ Faster time to market
- ✓ Better team efficiency
- ✓ Clear product vision within the team

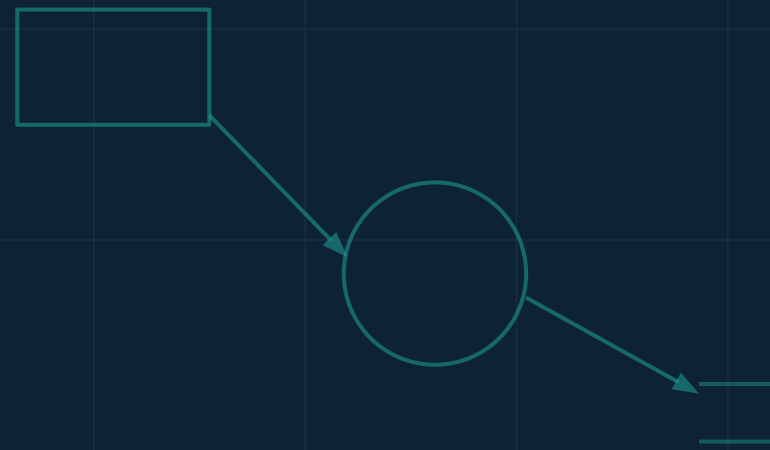
DevOps Challenges

- Difficulties with Integration
- Automated Testing
- Relatively High Costs
- Toolset Choice
- Lack of Talent

5.0

SEGMENT 05

Project Team Roles and Skills



SEGMENT 5

Information Systems Roles

BA

Business analyst

SA

System analyst

IA

Infrastructure analyst

CM

Change management analyst

PM

Project manager

Project Team Roles

ROLE	RESPONSIBILITIES
Business analyst	Analyzing the key business aspects of the system · Identifying how the system will provide business value · Designing the new business processes and policies
Systems analyst	Identifying how technology can improve business processes · Designing the new business processes · Designing the information system · Ensuring that the system conforms to information systems standards
Infrastructure analyst	Ensuring the system conforms to infrastructure standards · Identifying infrastructure changes needed to support the system
Change management analyst	Developing and executing a change management plan · Developing and executing a user training plan
Project manager	Managing the team of analysts, programmers, technical writers, and other specialists · Developing and monitoring the project plan · Assigning resources · Serving as the primary point of contact for the project

Summary — Part 1

The SDLC

The Systems Development Life Cycle consists of four stages:

1. Planning
2. Analysis
3. Design
4. Implementation

The major development methodologies

- Structured design
- The waterfall method
- Parallel development
- RAD development
- Prototyping (regular and throwaway)
- Agile development
- XP streamline SDLC
- DevOps

WRAP-UP

Summary — Part 2



There are five major team roles:

- **BA** — Business analyst
- **SA** — Systems analyst
- **IA** — Infrastructure analyst
- **CM** — Change mgmt. analyst
- **PM** — Project manager

End of Lecture 01 · Introduction & SDLC